Lab 2: Game Access Control on FPGA
User Manual

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1. Description

A game that can be played by two players on the DE0-CV FPGA board. Each player has one set of four slide switches on the board corresponding to a 4-bit binary number. Player 1 switches are the four rightmost switches (SW3-SW0) which SW0 is the least significant bit. Player 2 switches are the four leftmost switches (SW9-SW6) which SW6 is the least significant bit. Player 1 and Player 2' numbers are displayed on the rightmost 7-segment display and the second to leftmost display in hexadecimal value respectively. The sum of the two players' numbers is displayed on the third 7-segment display from the right in hexadecimal value. The leftmost LED is lit if the sum of the two numbers equal 4'b1111 or its right-adjacent LED is lit if otherwise.

Players have their own LOAD button that will display their respective number on the switches on the 7-segment displays. Before any player can load their number, the system password must be entered using Player 2's set of switches. If the password is successfully entered, the rightmost LED is lit up indicating logged-in status or the left-adjacent LED is lit if a correct password has not been entered. Once logged in and the password LOAD button is pressed again, players will be logged out. They cannot load their numbers until logged in again. The RESET button will log players out and set the board back to default settings.

The rule of the game is that Player 1 sets their number using the switches then Player 2 needs to set a number (one-time only) that the sum of both numbers equals 4'b1111 without looking at Player 1's switches. If Player 2 gets the right number, they win the round. If Player 2 gets the wrong number, Player 1 wins the round. The players swich roles after every round as Player 2 now sets a number for Player 1 to match.

Binary (switches)	Hexadecimal (display)	Decimal
0000	0	0
0001	1	1
0010	2	2
0011	3	3
0100	4	4
0101	5	5
0110	6	6
0111	7	7
1000	8	8
1001	9	9
1010	A	10
1011	B	11
1100	C	12
1101	D	13
1110	E	14
1111	F	15

Table 1: Hexadecimal (on display) and binary (switches) conversion to decimal

2. FPGA Board Overview

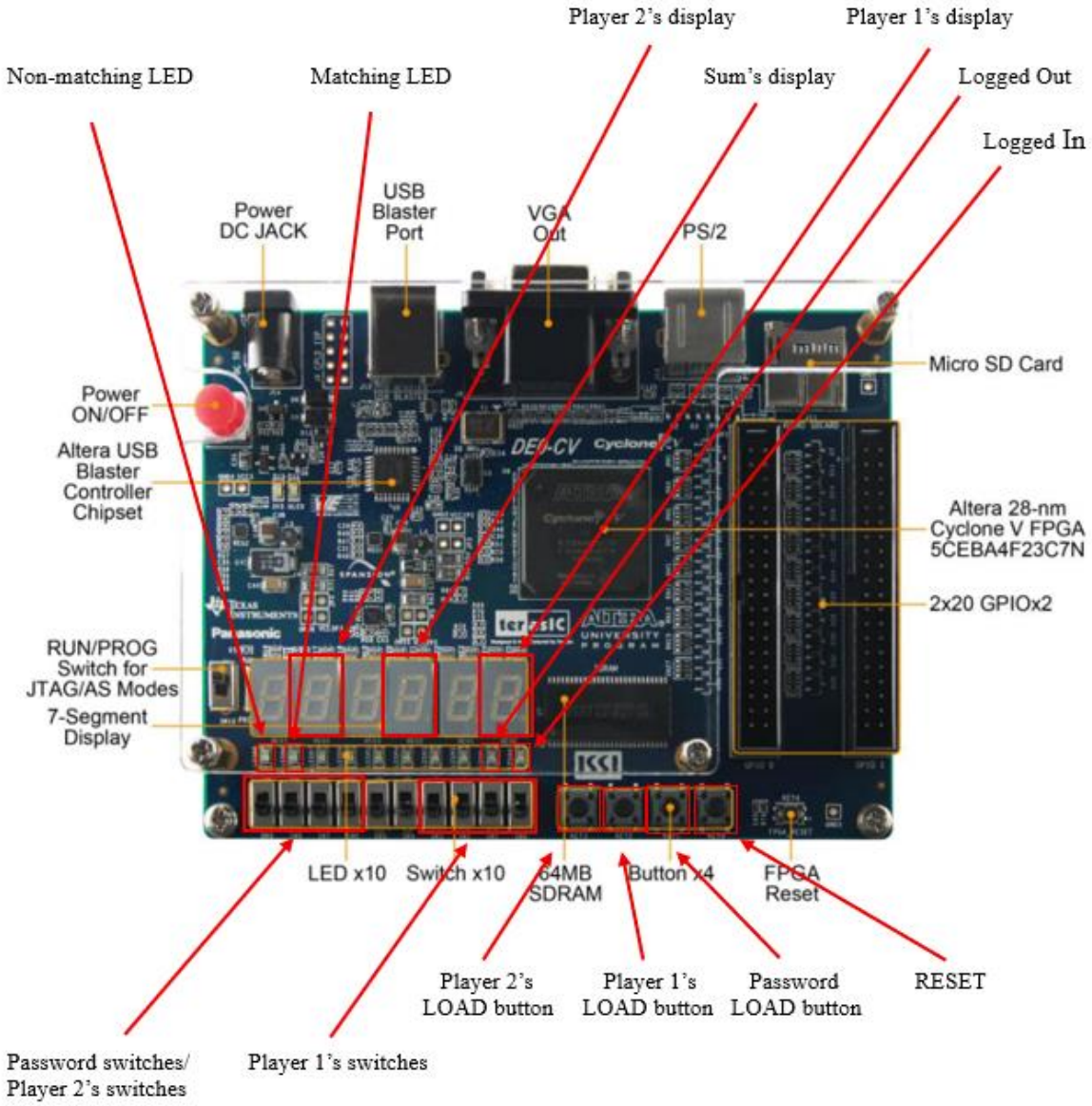

Figure 1: DE0-CV FPGA board with annotated Players' switches, displays, and LEDs

3. Step-by-step Game Rule

3.1. Authentication

Step 1: FPGA is powered and in default settings showing 0 on all three displays. The leftmost LED is lit depicting non-matching condition while the next-to-rightmost LED is lit indicating logged out status. The password is 1032.

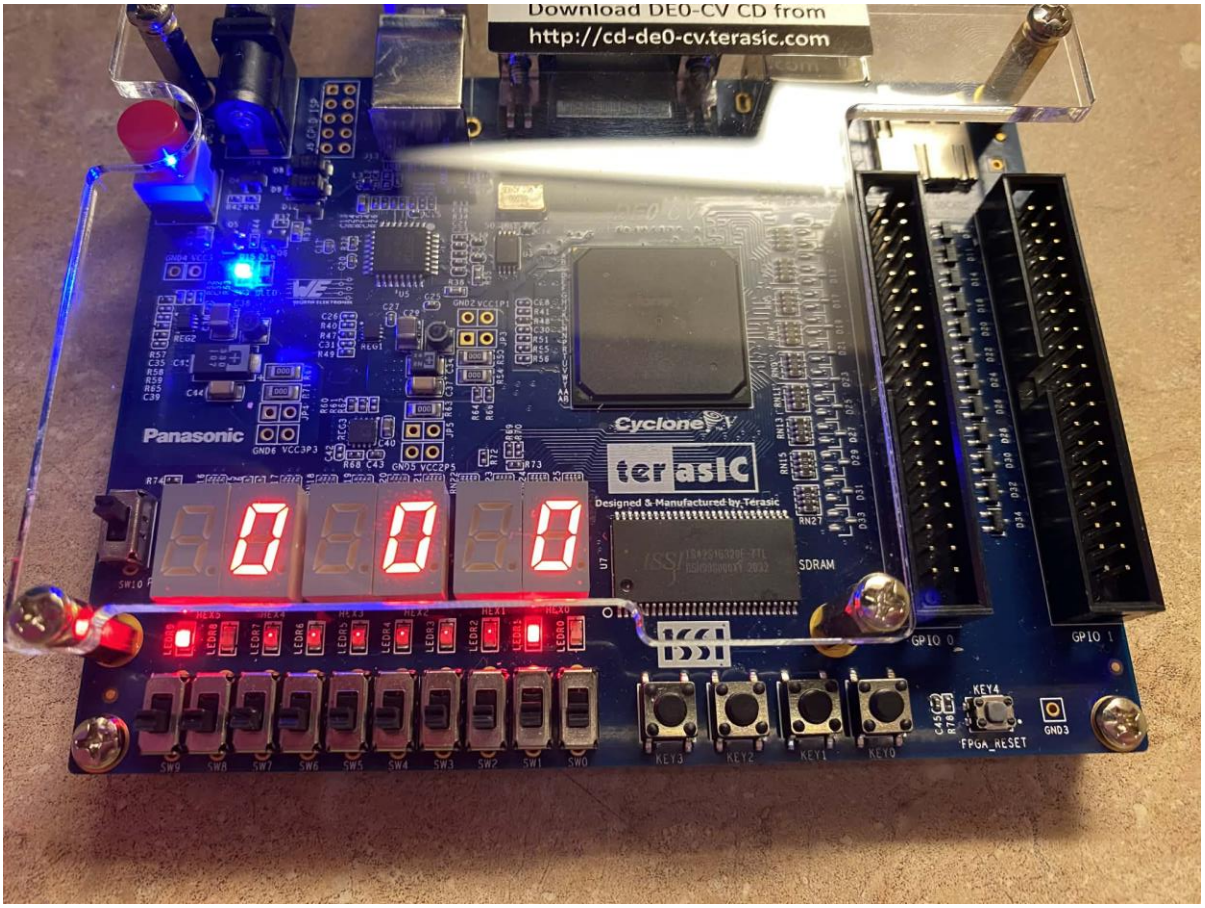

Figure 2: FPGA default settings

Step 2: The first password digit (1'd1 or 4'b0001) is entered using Player 2's switches. The same two LEDs are still lit.

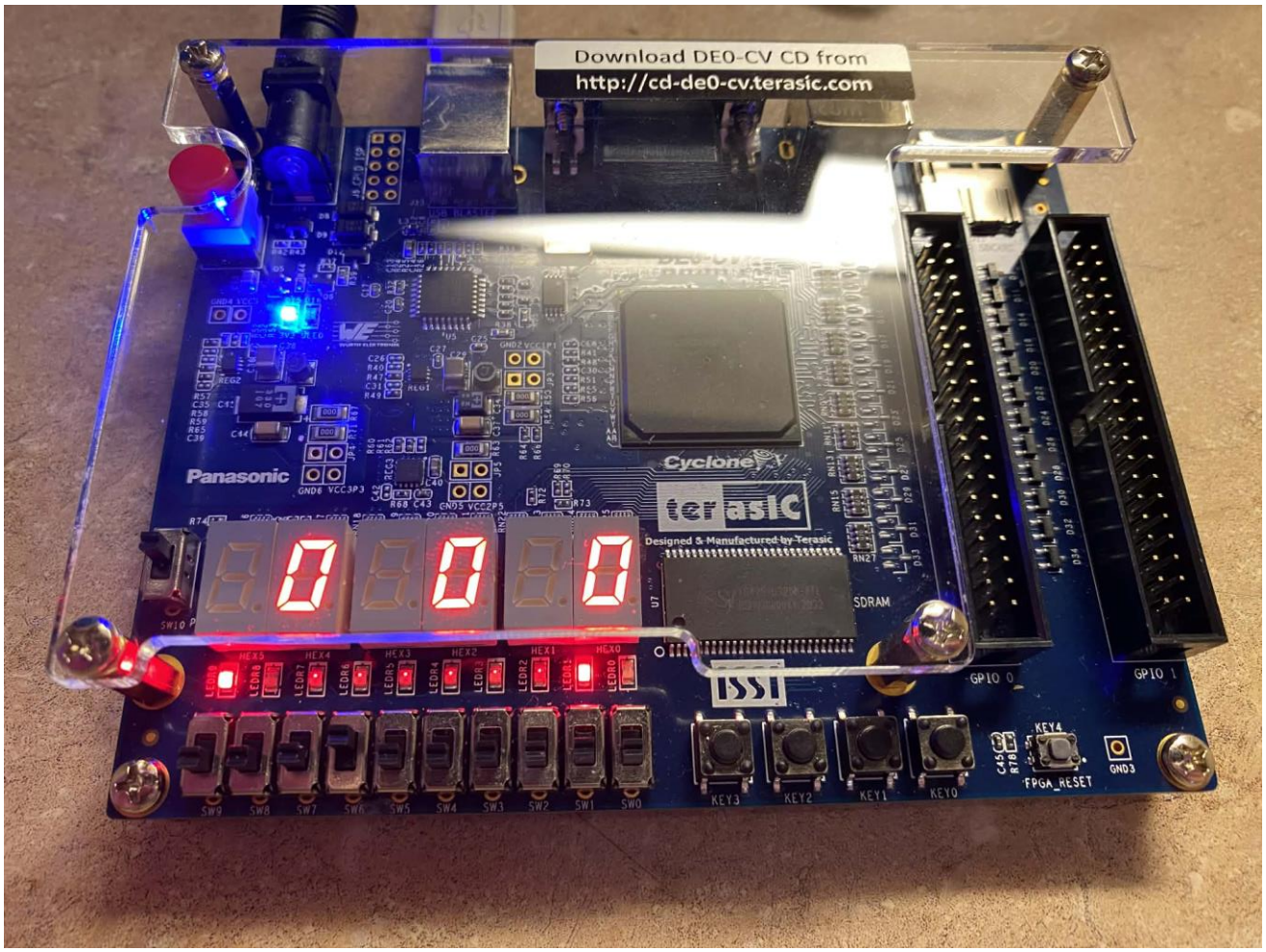

Figure 3: First password digit entered

Step 3: The second password digit (1'd0 or 4'b0000) is entered using Player 2's switches. The same two LEDs are still lit.

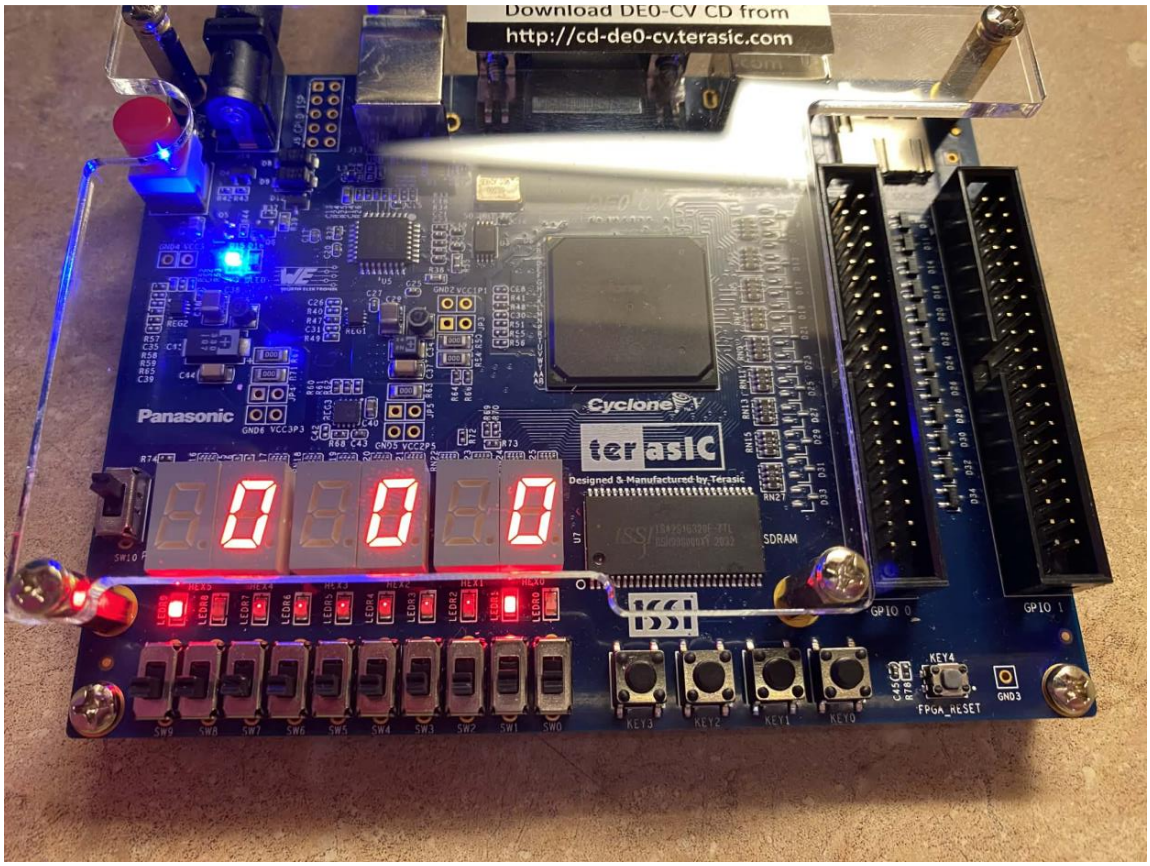

Figure 4: Second password digit entered

Step 4: The third password digit (1'd3 or 4'b0011) is entered using Player 2's switches. The same two LEDs are still lit.

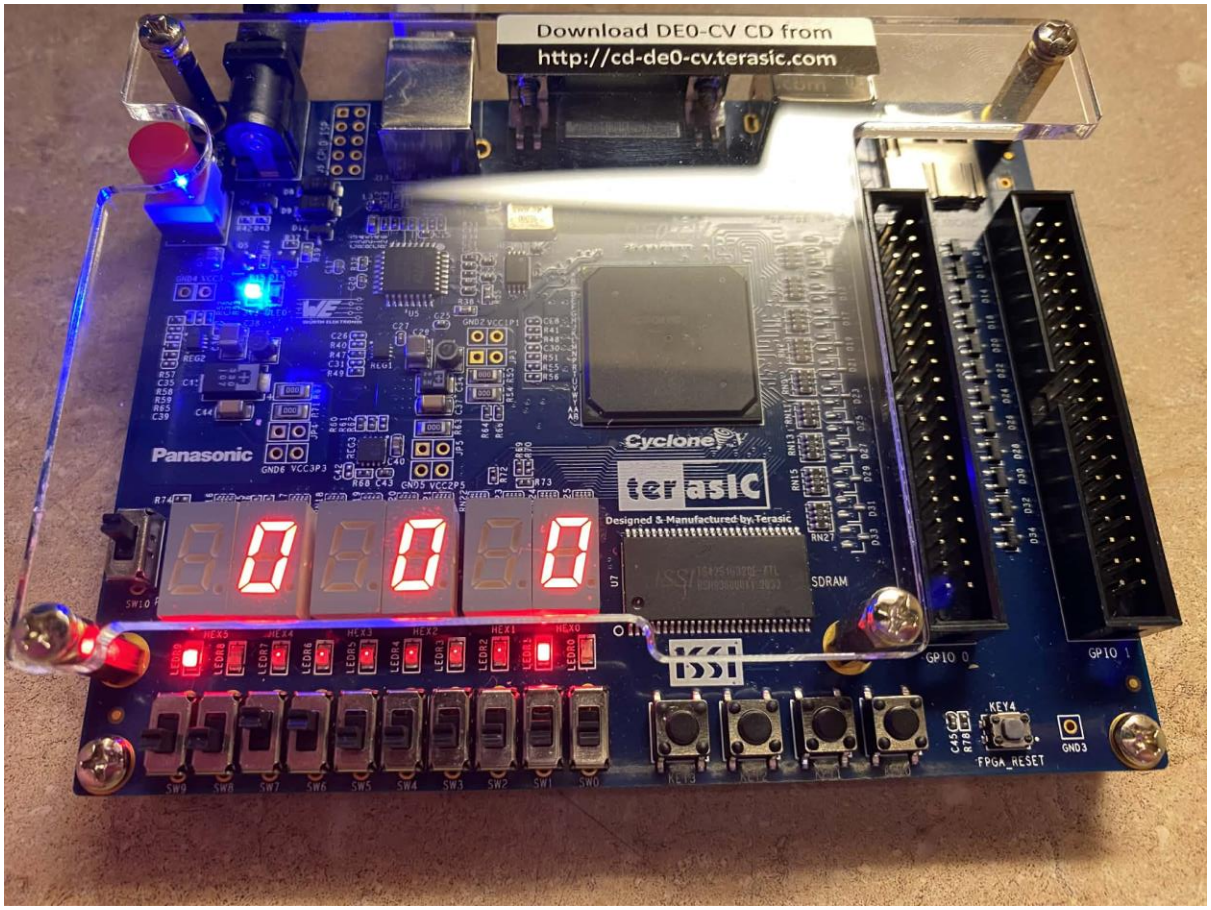

Figure 5: Third password digit entered

Step 5: The last password digit (1'd2 or 4'b0010) is entered using Player 2's switches. The same two LEDs are still lit.

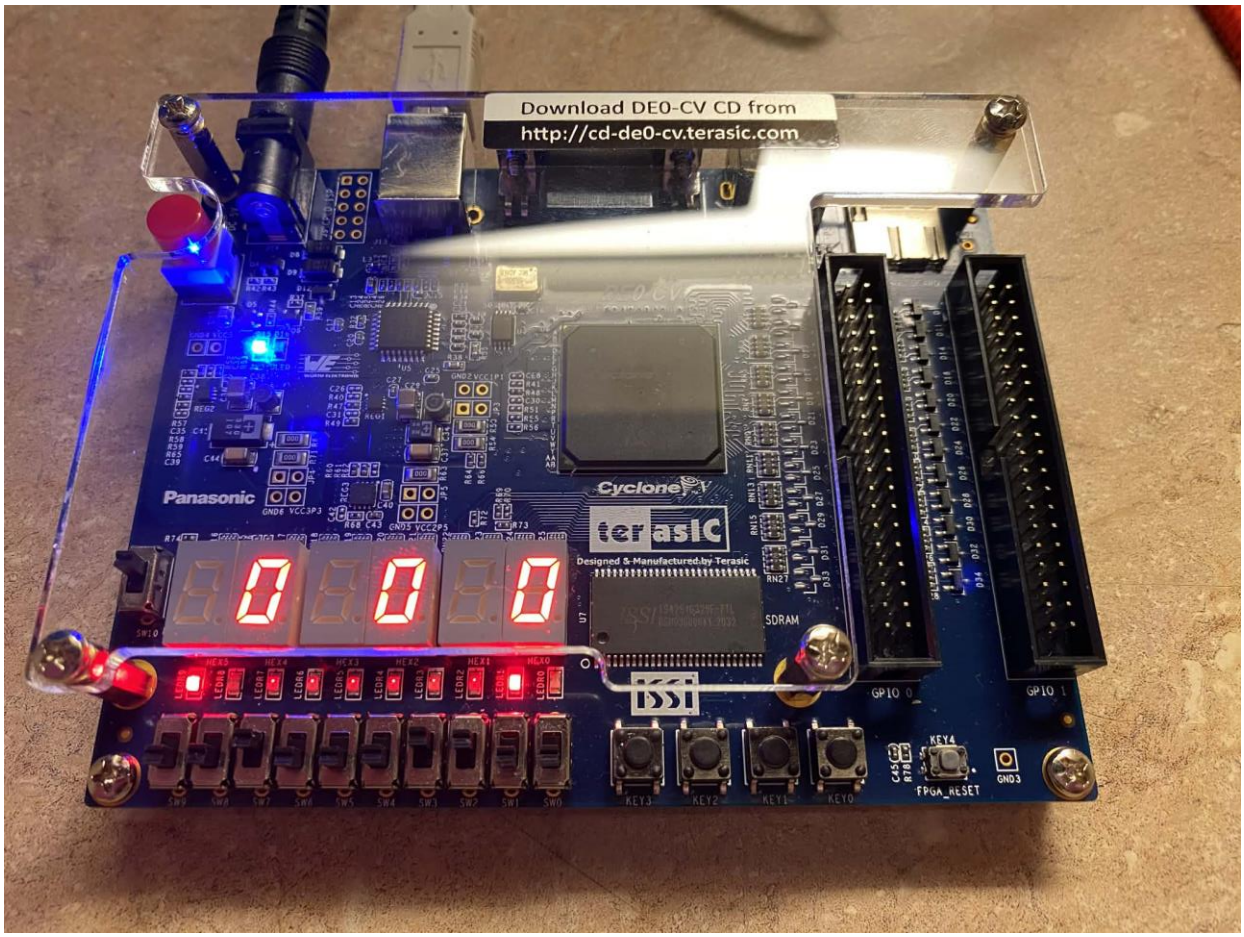

Figure 6: Last password digit entered

Step 6: After the last password digit is entered using Player 2's switches. The rightmost LED is lit indicating logged in status while the leftmost LED still shows non-matching condition because players haven't entered their numbers yet.

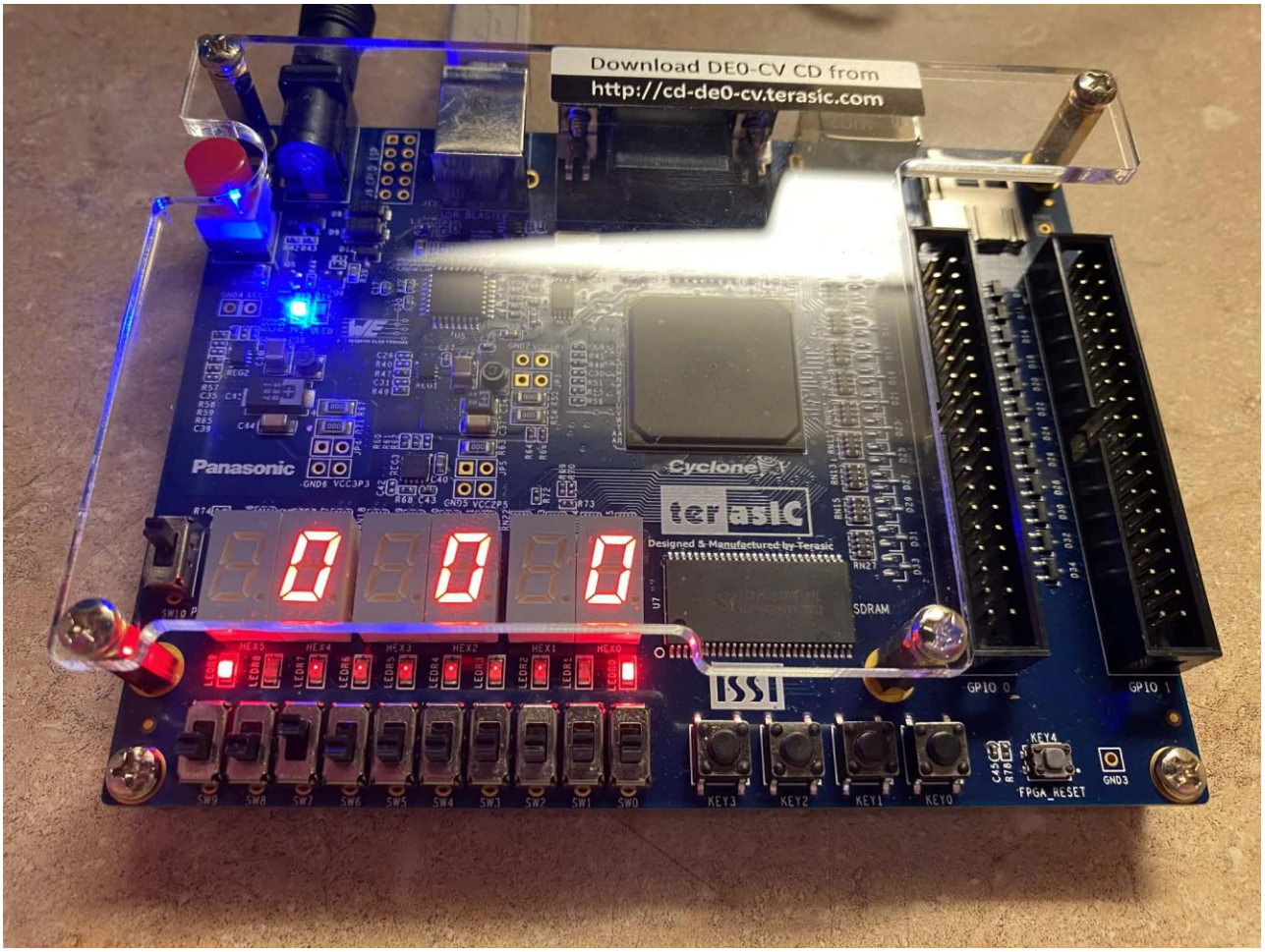

Figure 7: Players are logged in

3.2. Win condition

Step 1: Player 1 toggles their switches to enter 4'b1100 but haven't pushed their LOAD button yet. Therefore, the rightmost display still shows a 0.

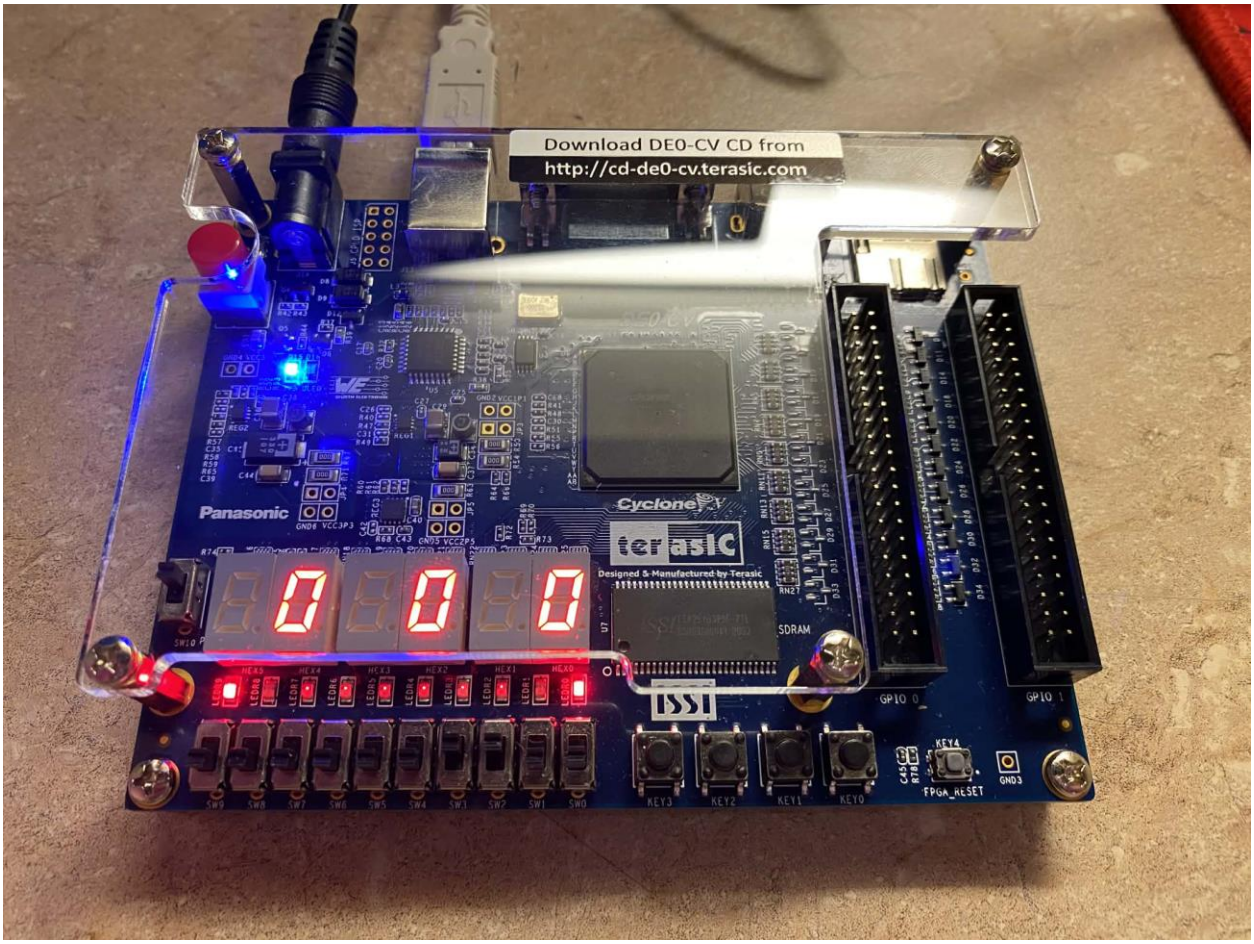

Figure 8: Player 1's number before loading

Step 2: Player 1 pushes their LOAD button and now their number is loaded. Player 1's display shows their number of 4'b1100 or 4'hC.

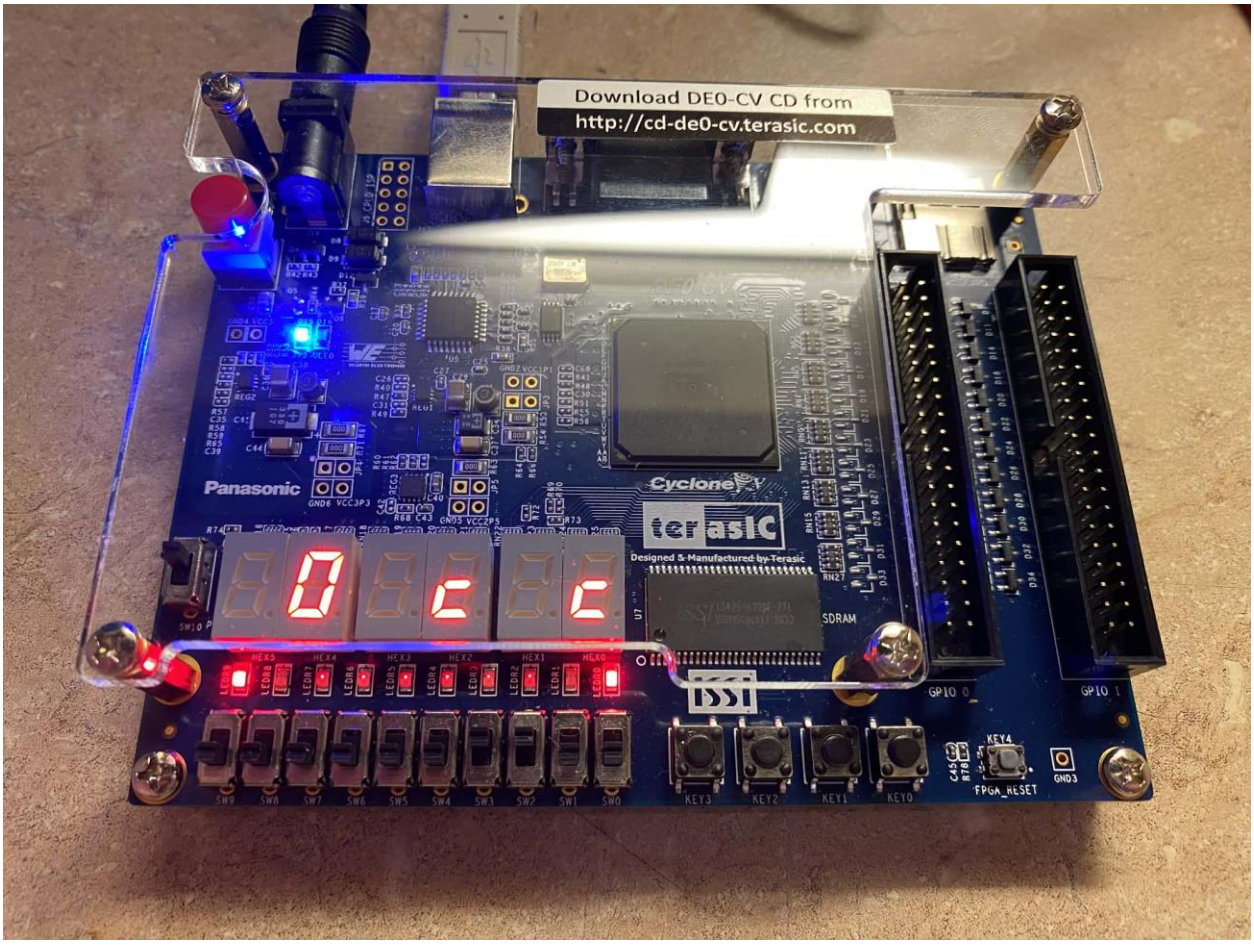

Figure 9: Player 1's number is loaded

Step 3: Player 2 toggles their switches to enter 4'b0011 but haven't pushed their LOAD button yet. Therefore, the next-to-leftmost display still shows a 0.

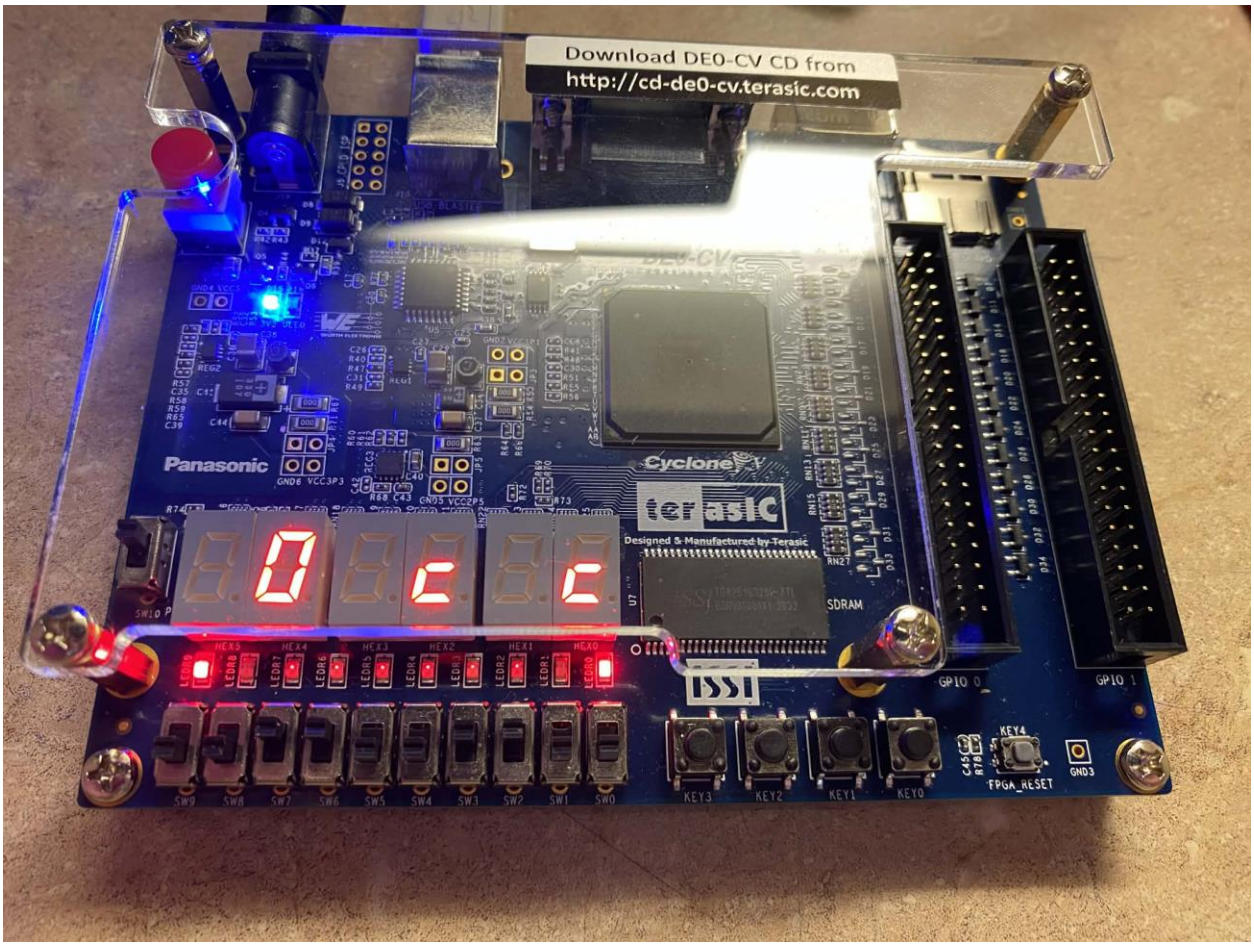

Figure 10: Player 2's number before loading

Step 4: Player 2 pushes their LOAD button and now their number is loaded. Player 2's display shows their number of 4'b0011 or 4'h3. The middle display also shows 4'hF. Now the next-to-leftmost LED is lit meaning the matching condition is met. This means Player 2 wins this round. In the next round, Player 1 will find the matching number.

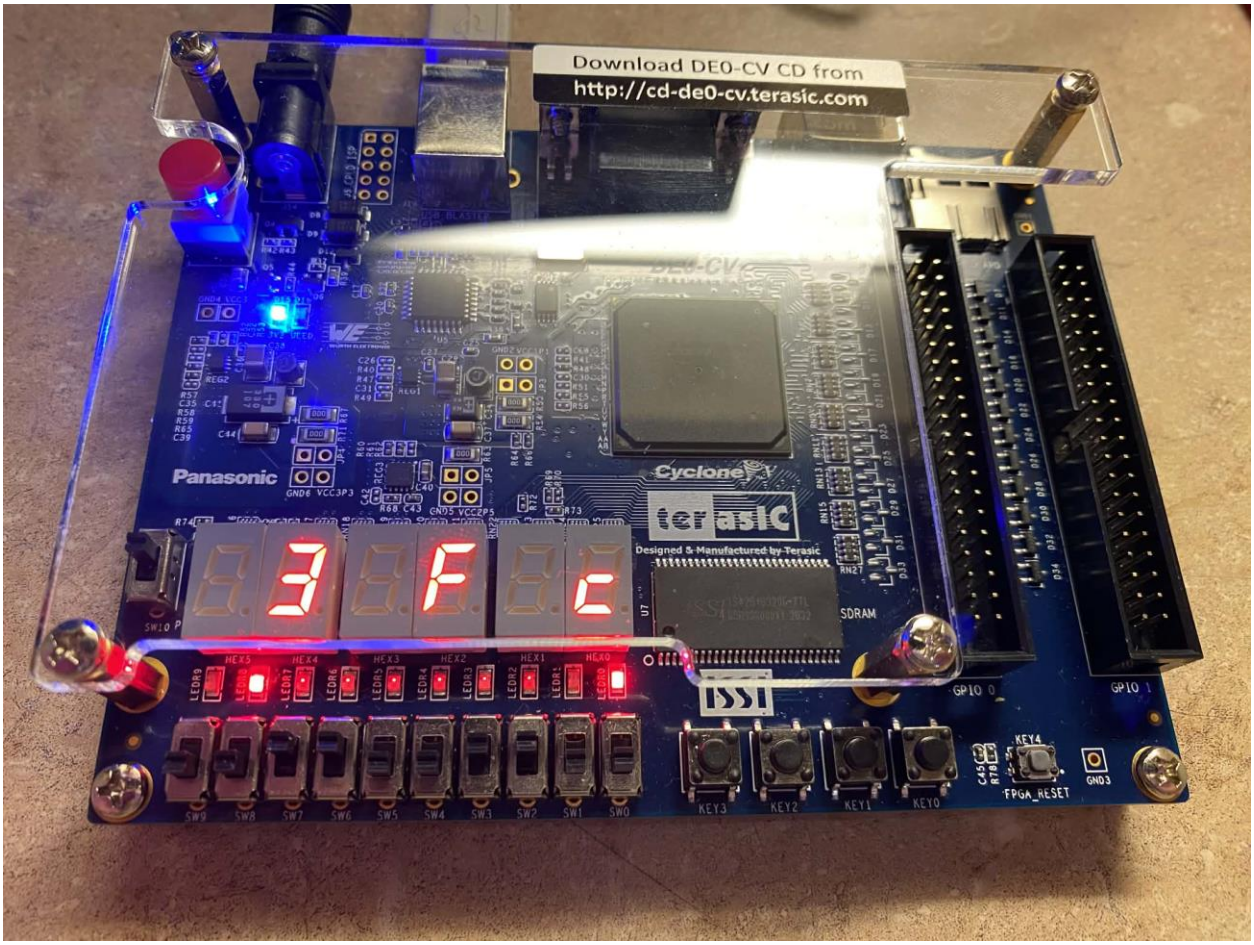

Figure 11: Player 2's number is loaded

3.3. Lose condition

The following figure shows a non-matching pair of numbers as the leftmost LED is lit up and the sum doesn't equal 4'hF. The player who found the non-matching number loses this round.

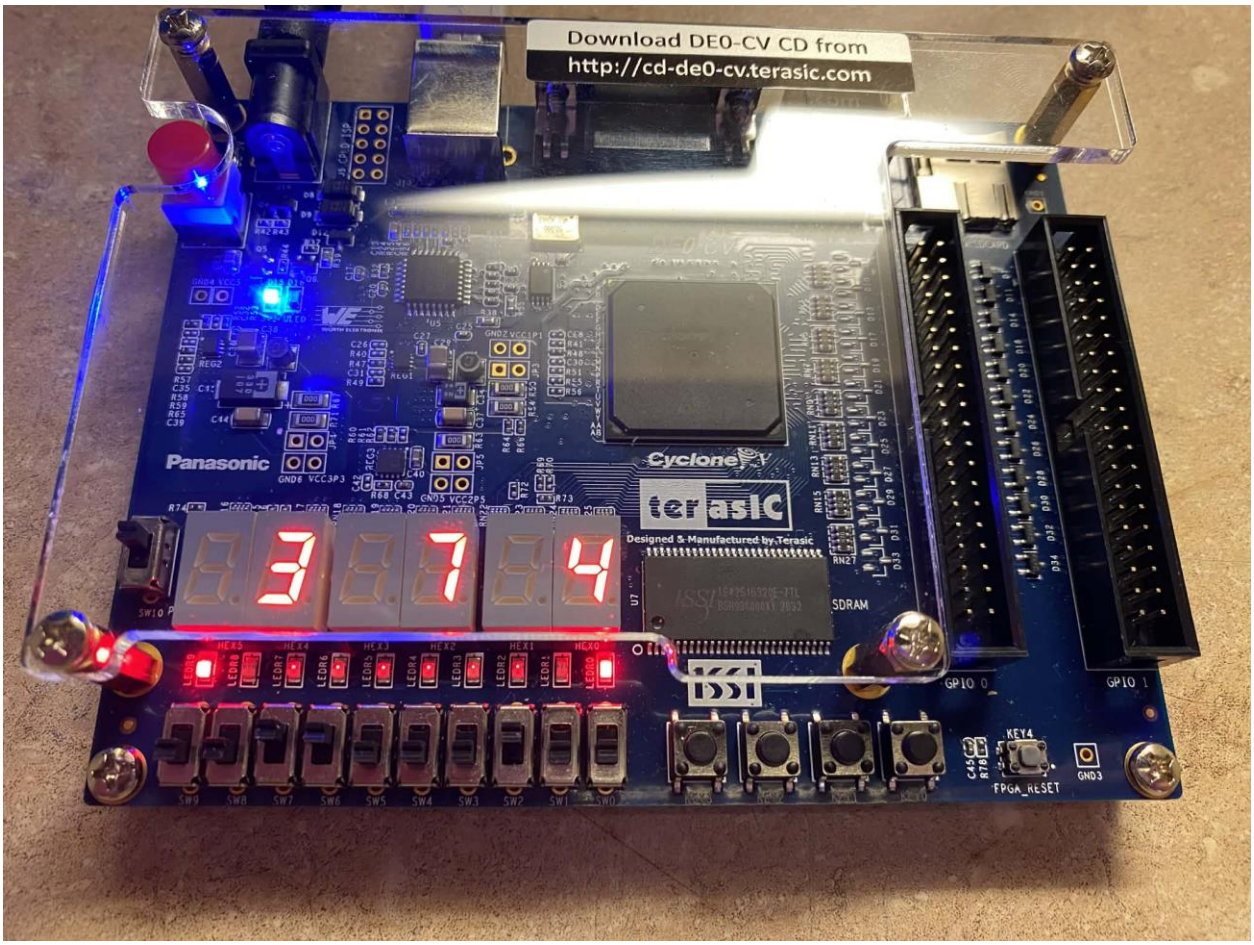

Figure 12: Non-matching condition

3.4.Log out function

Step 1: Continue from Step 4 of the matching condition, the password load button is pressed again. Now players are logged out again as the next-to-rightmost LED is lit indicating logged out status while the players' input numbers are maintained. If reset button is pushed, the numbers are also cleared.

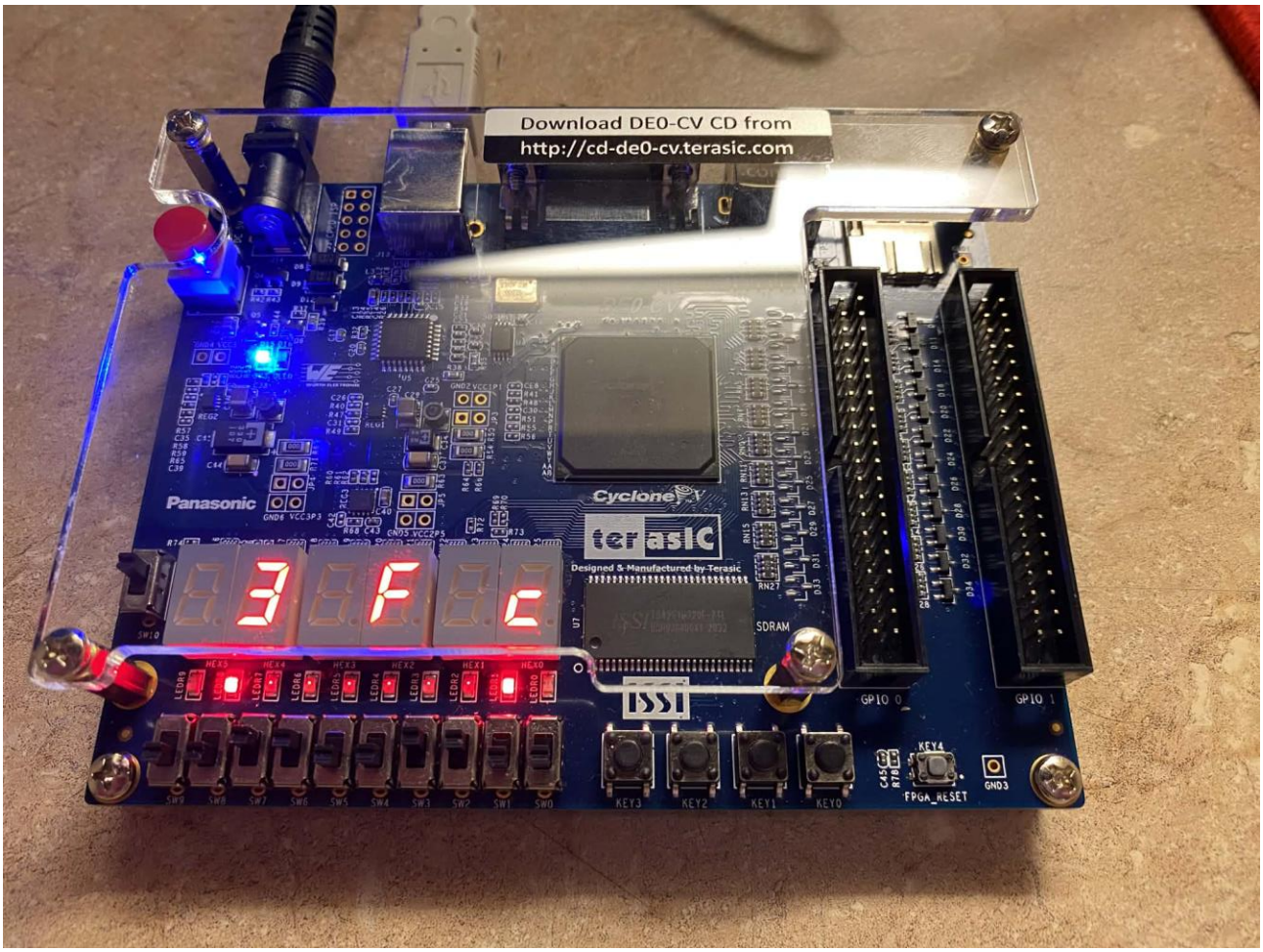

Figure 13: Log out function

Step 2: After successfully entering the password again, now players are logged back in as the rightmost LED is lit again.

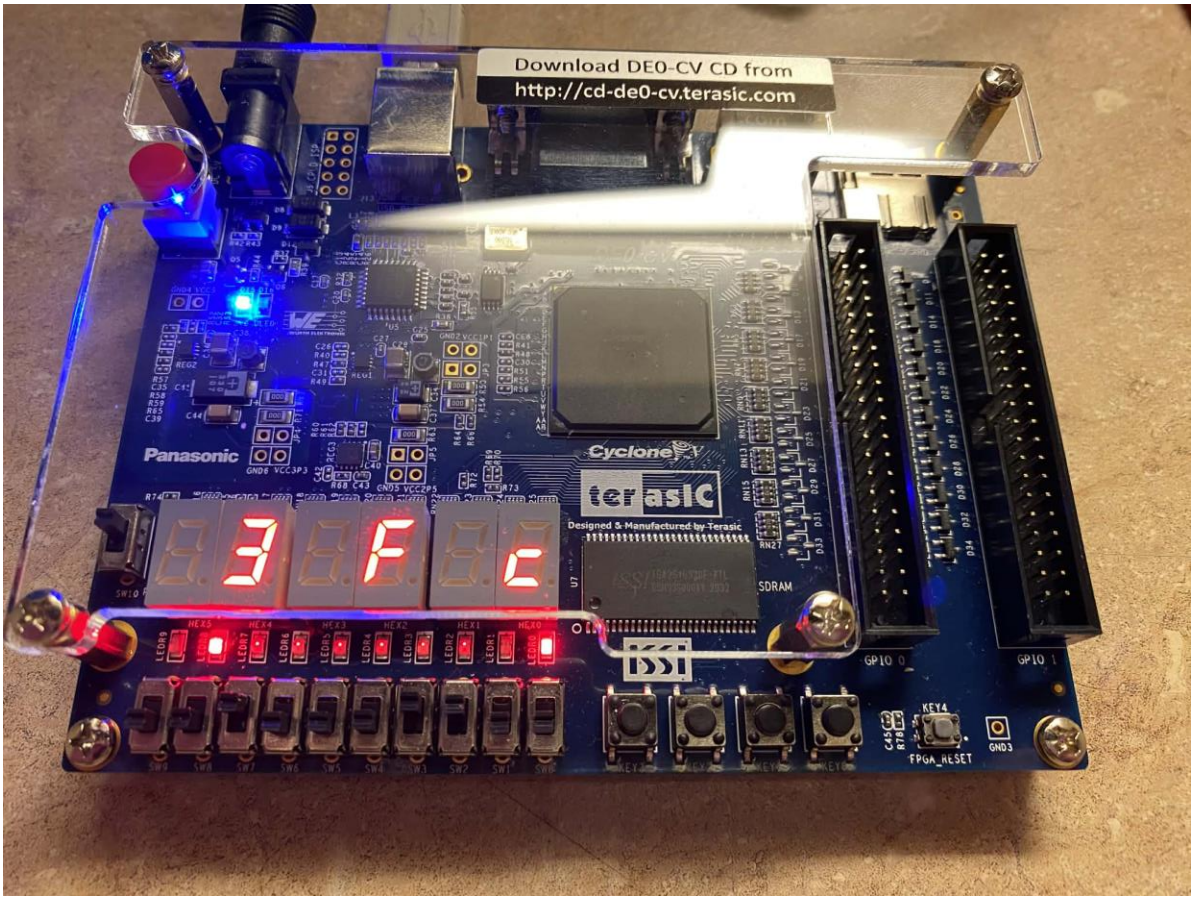

Figure 14: Logged back in